Department of Mathematics and Statistics, School of Arts and Science, Qatar University.

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| Unit Name: | Mathematics for EE | Instructor: | Dr. Mohamed Mabrok |
| Section: | L51 | Location: | Lecture Room |
| Exam : | **Fall 2024, Final Exam** | Time and Date: | December 9, 2024 |

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|---|---|
| Student Name: | |
| Student ID: | |

- No credit will be given to answers for non-requested questions.
- Only non-plotting calculators are permitted. The use of the cell phones is absolutely prohibited.
- Shut off your cell phone completely.
- Make sure that you are feeling comfortable before entering the class for the test.
- Full mark will be given for complete answers for all questions.
- Cheating is an academic violation according to Qatar University rules and regulations, and in some cases, it may result in final dismissal from the University.
- Students should not under any circumstances commit or participate in any cheating attempt or any act that violates student code of conduct.

| Question Number | Student Marks | Maximum Marks |
|---|---|---|
| **1** | | **7** |
| **2** | | **7** |
| **3** | | **16** |
| **4** | | **12** |
| **5** | | **15** |
| **6** | | **17** |
| **Total / (Max)** | | **74** |

1. Solve the following differential equation:

$$(2x + y\sin(y))dx + (xy\cos(y) + x\sin(y))dy = 0.$$

7 Marks

2. Solve the following differential equation:

$$x^2y'' + 5xy' + 3y = 0,$$

7 Marks

3. Compute the following:

$$A)\ \mathscr{L}^{-1}\left(\frac{3s+5}{s^2+s+1}\right)$$

$$B)\ \mathscr{L}\left(f(t)\right), \quad \text{where} \quad f(t)=\begin{cases} t & \text{if } t<1 \\ -e^{2t}+\left(t^2-1\right)^2 & \text{if } t\geq 1 \end{cases}$$

$$C)\ \mathscr{L}^{-1}\left(\frac{e^{-s}}{3s^2+12}\right)$$

$$D)\ \mathscr{L}\left(t^2\sin(2t)\right)$$

16 Marks

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4. Use Laplace transforms, find the general formula for the solution of the initial value problem for any $f(t)$.

$$y'' + 3y' = f(t), \quad y(0) = 0, \quad y'(0) = 0$$

Then, use the obtained general form and

$$f(t) = t$$

to find the solution (*You have to compute the integration*). 12 Marks

5. Show how the equation

$$y'' + 2y' + 5y = 0$$

can be transformed to the system

$$X' = \begin{bmatrix} 0 & 1 \\ -5 & -2 \end{bmatrix} X$$

Then, solve the system using matrix method. 15 Marks

6. Solve the following:

(a) Calculate the real and imaginary part of the following:

$$\frac{(3-i)^2-(1+3i)^2}{2i+2}$$

(b) Suppose that:

$$z_1 = 1+2i, \quad z_2 = -3-3i, \quad z_3 = 2i+4.$$

Write $z_1$ and $z_2$ in its polar form and compute $z_1\,\bar{z_3}$

(c) Find the all roots of $w^4 = 1+i$

(d) Show that $f(z) = 2\bar{z}$ is not differentible

17 Marks

## Algebra

- $\sin\theta = \frac{1}{\csc\theta}$
- $\csc\theta = \frac{1}{\sin\theta}$
- $\cos\theta = \frac{1}{\sec\theta}$
- $\sec\theta = \frac{1}{\cos\theta}$
- $\tan\theta = \frac{1}{\cot\theta}$
- $\cot\theta = \frac{1}{\tan\theta}$
- $\sin^2\theta + \cos^2\theta = 1$
- $1 + \tan^2\theta = \sec^2\theta$
- $1 + \cot^2\theta = \csc^2\theta$
- $\cos^2(x) = \frac{1}{2}(1 + \cos(2x))$
- $\sin^2(x) = \frac{1}{2}(1 - \cos(2x))$
- $\cos^2(x) = 1 - \sin^2(x)$
- $\sin^2(x) = 1 - \cos^2(x)$

## Differentiation

- $\frac{\mathrm{d}}{\mathrm{d}x} c = 0$
- $\frac{\mathrm{d}}{\mathrm{d}x} x^n = n x^{n-1}$
- $\frac{\mathrm{d}}{\mathrm{d}x} (ax \pm b)^n = a\, n\, (ax \pm b)^{n-1}$
- $\frac{\mathrm{d}}{\mathrm{d}x} \ln|x| = \frac{1}{x}$
- $\frac{\mathrm{d}}{\mathrm{d}x} \ln|ax \pm b| = \frac{a}{ax \pm b}$
- $\frac{\mathrm{d}}{\mathrm{d}x} e^{ax} = a\, e^{ax}$
- $\frac{\mathrm{d}}{\mathrm{d}x} \sin(ax) = a\, \cos(ax)$
- $\frac{\mathrm{d}}{\mathrm{d}x} \cos(ax) = -a\, \sin(ax)$
- $\frac{\mathrm{d}}{\mathrm{d}x} f(x)\, g(x) = f(x) \cdot \frac{\mathrm{d}g(x)}{\mathrm{d}x} + g(x) \cdot \frac{\mathrm{d}f(x)}{\mathrm{d}x}$ *Product rule.*
- $\frac{\mathrm{d}}{\mathrm{d}x} \frac{f(x)}{g(x)} = \frac{g(x) \cdot \frac{\mathrm{d}f(x)}{\mathrm{d}x} - f(x) \cdot \frac{\mathrm{d}g(x)}{\mathrm{d}x}}{(g(x))^2}$ *Quotient rule.*
- $\frac{\mathrm{d}}{\mathrm{d}x} f(g(x)) = f'(g(x)) \cdot g'(x)$ *Chain rule.*

## Integration

- $\int a\, dx = ax + c$
- $\int x^n\, dx = \frac{x^{n+1}}{n+1} + c \qquad n \neq -1$
- $\int f(x)^n f'(x)\, dx = \frac{f(x)^{n+1}}{n+1} + c \qquad n \neq -1$
- $\int \frac{a}{x}\, dx = a\, \ln|x| + c$
- $\int \frac{1}{ax+b}\, dx = \frac{1}{a} \ln|ax+b| + c$
- $\int e^{ax}\, dx = \frac{1}{a}\, e^{ax} + c$
- $\int \frac{f'(x)}{f(x)}\, dx = \ln|f(x)| + c$
- $\int \sin(ax)\, dx = \frac{-1}{a}\, \cos(ax) + c$
- $\int \cos(ax)\, dx = \frac{1}{a}\, \sin(ax) + c$
- $\int \sec^2(ax)\, dx = \frac{1}{a}\, \tan(ax) + c$
- $\int \csc^2(ax)\, dx = -\frac{1}{a}\, \cot(ax) + c$
- $\int \tan(ax)\, dx = -\frac{1}{a}\, \ln(\cos(ax)) + c$
- $\int \cot(ax)\, dx = \frac{1}{a}\, \ln(\sin(ax)) + c$
- $\int \sec(ax)\tan(ax)\, dx = \frac{1}{a}\, \sec(ax) + c$
- $\int \csc(ax)\cot(ax)\, dx = -\frac{1}{a}\, \csc(ax) + c$
- $\int \sec(ax) dx = \frac{1}{a}\, \ln(\sec(ax) + \tan(ax)) + c$
- $\int \csc(ax)\, dx = \frac{-1}{a}\, \ln(\csc(ax) - \cot(ax)) + c$
- $\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1}\left(\frac{x}{a}\right) + C$
- $\int \frac{1}{x^2 + a^2} dx = \frac{1}{a} \tan^{-1}\left(\frac{x}{a}\right) + C$
- $\int \frac{1}{x\sqrt{x^2 - a^2}} dx = \frac{1}{a} \sec^{-1}\left(\frac{x}{a}\right) + C$

## Table of Laplace Transforms

| | $f(t)=\mathfrak{L}^{-1}\{F(s)\}$ | $F(s)=\mathfrak{L}\{f(t)\}$ | | $f(t)=\mathfrak{L}^{-1}\{F(s)\}$ | $F(s)=\mathfrak{L}\{f(t)\}$ |
|---|---|---|---|---|---|
| 1. | $1$ | $\frac{1}{s}$ | 2. | $\mathbf{e}^{at}$ | $\frac{1}{s-a}$ |
| 3. | $t^n,\quad n=1,2,3,\ldots$ | $\frac{n!}{s^{n+1}}$ | 4. | $t^p,\ p>-1$ | $\frac{\Gamma(p+1)}{s^{p+1}}$ |
| 5. | $\sqrt{t}$ | $\frac{\sqrt{\pi}}{2s^{\frac{3}{2}}}$ | 6. | $t^{n-\frac{1}{2}},\quad n=1,2,3,\ldots$ | $\frac{1\cdot 3\cdot 5\cdots(2n-1)\sqrt{\pi}}{2^n s^{n+\frac{1}{2}}}$ |
| 7. | $\sin(at)$ | $\frac{a}{s^2+a^2}$ | 8. | $\cos(at)$ | $\frac{s}{s^2+a^2}$ |
| 9. | $t\sin(at)$ | $\frac{2as}{(s^2+a^2)^2}$ | 10. | $t\cos(at)$ | $\frac{s^2-a^2}{(s^2+a^2)^2}$ |
| 11. | $\sin(at)-at\cos(at)$ | $\frac{2a^3}{(s^2+a^2)^2}$ | 12. | $\sin(at)+at\cos(at)$ | $\frac{2as^2}{(s^2+a^2)^2}$ |
| 13. | $\cos(at)-at\sin(at)$ | $\frac{s(s^2-a^2)}{(s^2+a^2)^2}$ | 14. | $\cos(at)+at\sin(at)$ | $\frac{s(s^2+3a^2)}{(s^2+a^2)^2}$ |
| 15. | $\sin(at+b)$ | $\frac{s\sin(b)+a\cos(b)}{s^2+a^2}$ | 16. | $\cos(at+b)$ | $\frac{s\cos(b)-a\sin(b)}{s^2+a^2}$ |
| 17. | $\sinh(at)$ | $\frac{a}{s^2-a^2}$ | 18. | $\cosh(at)$ | $\frac{s}{s^2-a^2}$ |
| 19. | $\mathbf{e}^{at}\sin(bt)$ | $\frac{b}{(s-a)^2+b^2}$ | 20. | $\mathbf{e}^{at}\cos(bt)$ | $\frac{s-a}{(s-a)^2+b^2}$ |
| 21. | $\mathbf{e}^{at}\sinh(bt)$ | $\frac{b}{(s-a)^2-b^2}$ | 22. | $\mathbf{e}^{at}\cosh(bt)$ | $\frac{s-a}{(s-a)^2-b^2}$ |
| 23. | $t^n\mathbf{e}^{at},\quad n=1,2,3,\ldots$ | $\frac{n!}{(s-a)^{n+1}}$ | 24. | $f(ct)$ | $\frac{1}{c}F\left(\frac{s}{c}\right)$ |
| 25. | $u_c(t)=u(t-c)$ Heaviside Function | $\frac{\mathbf{e}^{-cs}}{s}$ | 26. | $\delta(t-c)$ Dirac Delta Function | $\mathbf{e}^{-cs}$ |
| 27. | $u_c(t)f(t-c)$ | $\mathbf{e}^{-cs}F(s)$ | 28. | $u_c(t)g(t)$ | $\mathbf{e}^{-cs}\mathfrak{L}\{g(t+c)\}$ |
| 29. | $\mathbf{e}^{ct}f(t)$ | $F(s-c)$ | 30. | $t^n f(t),\quad n=1,2,3,\ldots$ | $(-1)^n F^{(n)}(s)$ |
| 31. | $\frac{1}{t}f(t)$ | $\int_s^\infty F(u)\,du$ | 32. | $\int_0^t f(v)\,dv$ | $\frac{F(s)}{s}$ |
| 33. | $\int_0^t f(t-\tau)g(\tau)\,d\tau$ | $F(s)G(s)$ | 34. | $f(t+T)=f(t)$ | $\frac{\int_0^T \mathbf{e}^{-st}f(t)\,dt}{1-\mathbf{e}^{-sT}}$ |
| 35. | $f'(t)$ | $sF(s)-f(0)$ | 36. | $f''(t)$ | $s^2F(s)-sf(0)-f'(0)$ |
| 37. | $f^{(n)}(t)$ | $s^nF(s)-s^{n-1}f(0)-s^{n-2}f'(0)\cdots-sf^{(n-2)}(0)-f^{(n-1)}(0)$ | | | |